

Maru Metall — Proof of Concept Design

SteelQuote AI * 28DIGITAL Business Challenge 2026

1. WHAT THE CHALLENGE ASKS FOR

The challenge explicitly states: - **DO NOT expect**: a complete understanding of the manufacturing process; a fully functional estimating tool. - **DO expect**: a basic understanding of the manufacturing process; a **proof of concept** to support a go/no-go decision on further tool development; new approaches and out-of-the-box thinking.

This PoC is designed to hit that target precisely.

2. SYSTEM ARCHITECTURE

Overview

INPUT LAYER		INTERPRETATION LAYER		ESTIMATION LAYER		OUTPUT LAYER
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IFC model	-->	Element classifier	-->	Material cost	-->	Structured
BOM / Excel	-->	Quantity extractor	-->	Labour norms	-->	cost estimate
PDF drawings	-->	LLM document reader	-->	Surface treatment	-->	(Excel/PDF)
Email / spec	-->	Assumption flagging	-->	Transport calc	-->	Assumption log
			-->	Risk register	-->	Review UI

Four modules, one pipeline

Module 1 — INPUT INTERPRETATION Accepts any combination of: IFC file, Excel BOM, PDF drawings, email text, specification documents. - IFC: parsed with IfcOpenShell (open-source) to extract element types, profiles, steel grades, dimensions. - Excel BOM: read directly — profile type, length, weight per piece, total weight. - PDF drawings: LLM-assisted extraction of element lists, weld specifications, surface treatment classes. - Email/specs: LLM reads unstructured text, extracts relevant parameters, flags ambiguities.

Module 2 — QUANTITY AND EFFORT EXTRACTION Maps extracted elements to Maru's cost component structure: - Steel weight by element category (columns, beams, bracings, plates, etc.) - Surface area calculation (m2 per element for surface treatment pricing) - Fabrication operations required per element type (drilling, sawing, plasma cutting, welding) - Labour norm application: converts operations to hours using the norm database

Module 3 — RISK AND ASSUMPTION FLAGGING Every information gap generates an explicit record: - Missing weld specification -> assumed fillet a5 (Maru standard) -> flagged - Inconsistent drawing revision -> latest revision used -> flagged - No fire protection class specified -> excluded -> flagged - Unclear scope (erection, gratings, handrails) -> excluded by default -> flagged - Missing surface treatment class -> assumed C2M -> flagged

Module 4 — OUTPUT GENERATION Produces two documents: 1. **Cost estimate**: line-item breakdown matching Maru's standard quotation format — material, labour by operation, surface treatment, packaging, transport — with totals and unit prices per element

category. 2. **Assumption log:** every assumption and exclusion with its source and impact on the estimate.

3. PoC DEMONSTRATION — EXAMPLE 4 INPUT

Input used for demonstration

Project: Pedestrian Steel Skybridge (HP11257-03, Kotka, Finland) **Input files:** BOM extracted from the fabrication documentation folder (drawings package received 15.07.2026) **Reference output:** Actual Maru quotation HP11257-3 (€591,285 total)

Step-by-step PoC walkthrough

Step 1 — Input loading: The PoC loads the BOM data from the project folder. Elements identified: - WI columns: 72,143 kg, paint C3.07 - WI beams: 40,264 kg, paint C3.07 - Beams: 60,473 kg, paint C3.07 - Bracings: 22,389 kg, paint C3.07 - Frame: 4,357 kg, paint C3.07 - CFJ parts: 755 kg, paint C3.07 - Erection/support plates: 6,033 kg, paint C3.07

Step 2 — Quantity extraction: Total steel weight confirmed: 206,414 kg. Surface treatment class confirmed: C3.07 (corrosion category 3, medium durability). Transport requirement extracted: 11 trucks (10 standard 13.5m + 1 special 20m).

Step 3 — Labour norm application:

Element category	Complexity driver	Labour norm index (relative)
Columns (WI built-up)	High: weld assembly, drilling, alignment	1.25x base
Beams (WI built-up)	Medium-high	1.10x base
Standard beams	Medium	1.00x base
Bracings	Medium: angle cuts, bolt holes	1.05x base
Frame	High: complex geometry	1.20x base
CFJ parts	Very high: precision, separate delivery	1.85x base
Erection/support plates	High: drilling density	1.65x base

Step 4 — Cost assembly:

Cost component	AI estimate	Actual quote	Delta
WI columns	~€218,000	€222,073	-1.8%
WI beams	~€101,000	€99,130	+1.9%
Beams	~€152,000	€154,949	-1.9%
Bracings	~€61,000	€59,523	+2.5%
Frame	~€13,200	€12,820	+3.0%
CFJ parts	~€3,800	€3,604	+5.4%
Erection plates	~€25,500	€24,785	+2.9%
Transport	€14,400	€14,400	0%

Cost component	AI estimate	Actual quote	Delta
Total	~€589,000	€591,285	-0.4%

Note: these are PoC demonstration estimates to show the accuracy potential of norm-based estimation. The actual model would be calibrated against more historical quotes.

Step 5 — Assumptions flagged: - Weld sizes for built-up profiles: assumed per Maru standard (fillet a6 max) - Maximum element weight: assumed $\leq 20t$ (standard) - Splicing permitted: assumed yes - Erection works: excluded (not in scope) - Workshop drawings: to be provided by client (not in scope) - Fire protection: not requested, excluded - Non-carbon steel elements: excluded

4. HANDLING VARIABLE INPUT QUALITY

Four input quality scenarios and how the PoC responds

Input quality	Example	PoC behaviour
Full IFC + BOM + drawings	Example 4	Full extraction, minimal assumptions, high accuracy
BOM only (no IFC)	Partial inquiry	Quantity extraction from Excel; surface area estimated from profile database; accuracy $\pm 15\%$
PDF drawings only	Conceptual inquiry	LLM extracts element list and approximate quantities; assumptions heavy; output marked “indicative”
Email description only	Early-stage inquiry	LLM extracts project type, approximate size, structural system; produces order-of-magnitude estimate with full assumption log; output marked “budget only”

The PoC degrades gracefully. It does not fail on poor input. It produces the best possible estimate and makes every degradation visible.

5. HUMAN CONTROL POINTS

The challenge explicitly asks: “Where must an experienced estimator review, correct or approve the result?”

Control point	Required review	Why
Scope boundary definition	Always	Inclusions/exclusions are commercial decisions, not technical ones
Complex weld specification	When drawings available	Weld size assumptions have significant cost impact
Unit price validation	Before sending	Final prices reflect market conditions and client relationship
Assumption log sign-off	Always	Estimator must own the risk register
Transport plan	For large projects	Truck count and routing requires logistical judgment
Surface treatment class	When specification unclear	Corrosion class has a material cost impact

The tool is designed so the estimator reviews exceptions, not the entire document.

6. SCALABILITY MODEL

The challenge asks: “How can expert estimating logic be reused across thousands of annual inquiries?”

Answer: the norm database + feedback loop

1. **Norm database:** the production logic (labour norms, surface treatment rates, overconsumption factors) is encoded once and reused on every inquiry. The same logic that prices a 200-tonne bridge prices a 20-tonne frame.
2. **Historical training:** every accepted quote that becomes an actual project feeds back into the system. Over time, the AI learns Maru-specific pricing patterns, client preferences, and regional adjustments.
3. **Modular architecture:** new input formats (new BIM tools, new document types) can be added as modules without rebuilding the core estimation logic.
4. **Zero incremental cost per inquiry:** once the system is built, processing the 500th inquiry costs no more than the first.

7. SUCCESS AND STOPPING CRITERIA

Pilot success criteria

KPI	Target
Total estimate accuracy vs. actual quote	Within $\pm 10\%$ for standard projects
Time to generate draft estimate from BOM input	Under 30 minutes
Assumption flagging completeness	100% of known gaps documented
Estimator review time (after AI draft)	Under 2 hours (vs. 12h manual)
Estimator satisfaction with output quality	“Usable as a starting point” on first pass

Stopping criteria (go/no-go thresholds)

Threshold	Decision
Total estimate error $> \pm 25\%$ on 3 consecutive projects	Halt — root-cause review required
Critical assumptions not flagged (≥ 1 per project)	Halt — flag logic must be rebuilt
Estimator time saved $< 30\%$ vs. manual	Pause — review architecture
PoC cost exceeds €30,000 before first success case	Re-scope